

YUBO ZHANG

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EDUCATION BACKGROUND

Columbia University

- Ph.D. in Electrical Engineering
- Overall GPA: 4.0/4.3
- Advisor: Prof. Xiaodong Wang

New York, NY
Aug. 2024 - Now

University of Science and Technology of China (USTC)

- Master of Science in Information and Communication Engineering
- Overall GPA: 3.6/4.3
- Advisor: Prof. Chen Gong

Hefei, China
Sept. 2021 - Jul. 2024

University of Science and Technology of China (USTC)

- Bachelor of Engineering in Electronic Information Engineering
- Overall GPA: 3.7/4.3, ranking top 10% out of 313 students

Hefei, China
Sept. 2017 - Jul. 2021

SELECTED RESEARCH EXPERIENCE

Transformer for Scalable MU-MISO Beamforming Optimization

Apr. 2025 - Oct. 2025

Research Assistant, Columbia

- Designed and trained a multi-layer Transformer model from scratch for large-scale MU-MISO beamforming optimization, integrating curriculum learning and semi-amortized learning strategies.
- Implemented the full architecture and training pipeline in PyTorch. Achieved 5% higher sum-rate than the WMMSE algorithm with only 3% of its inference time, while maintaining strong generalization to dynamic communication scenarios.
- Employed random masked attention to enhance scalability and robustness, and planned trainable sparse attention for improved training efficiency.

Encoder-Decoder Network for FDD Large-Scale Sparse Channel Precoding

Sep. 2024 - Apr. 2025

Research Assistant, Columbia

- Designed and trained an encoder-decoder network for channel recovery and beamforming design under large-scale sparse channels, and extended the framework to hybrid beamforming under far-field and near-field scenarios.
- Completed the problem formulation, architecture, training algorithm, and PyTorch implementation. Addressed local-optima entrapment and feedback overhead; the model outperformed MMSE baselines by 13% - 37% across channel models and SNR settings.

Resource Allocation Optimization for Keyless Positive-Rate Covert Communication

Feb. 2024 - Aug. 2024

Research Assistant, Columbia

- Developed an MDP model and trained a deep Q-network tailored to causal allocation problems; proposed multi-step non-convex optimization schemes for non-causal allocation problems.
- Completed the optimization and DQN framework design and PyTorch implementation. The DQN approach improved covert-rate performance by 10% - 26% compared with average-allocation schemes.

Graph-Based Conflict-Free MAC Protocol for the Communication Network

Dec. 2022 - Jun. 2023

Research Assistant, USTC

- Designed a resource-allocation scheme for a two-layer network, analyzed link interference using a conflict graph and Monte-Carlo simulation, and optimized routing algorithms.
- Contributed to interference-model analysis, routing-algorithm design, and data-fitting work.

Multi-User Uplink Communications with Spatial-Multiplexing Receiver

Sept. 2022 - Mar. 2023

Research Assistant, USTC

- Designed pilot sequences using an analytical MSE objective, analyzed Gaussian approximation, and adopted successive-elimination detection to handle unknown interference.
- Implemented the estimation/detection schemes in MATLAB. The pilot design approached optimal estimation MSE, while SED achieved nearly ML-level BER with only 5% - 10% of the execution time.

SELECTED PROJECT EXPERIENCE

1Mbps Multi-User Ad-hoc Ultraviolet Optical Communication System on FPGA

Apr. 2023 - Sept. 2023

Hardware Engineer, Software Engineer

- Achieved network synchronization using beacon sequences, designed the frame structure, and developed a multi-transceiver TDMA communication network.
- Integrated physical- and MAC-layer modules, conducted system-level testing, and completed FPGA development/debugging of synchronization and TDMA modules.

TDOA-Based Ultraviolet Positioning System on FPGA

May 2021 - Sept. 2021

Hardware Engineer

- Developed a TDOA-based ultraviolet positioning system and conducted large-scale outdoor field tests.
- Responsible for system-level code implementation, data acquisition, and performance validation.

10Kbps Multi-Service Real-Time Ultraviolet Optical Communication System on FPGA

May 2021 - Sept. 2021

Software Engineer, Test Engineer

- Developed an FPGA-based ultraviolet optical communication system, designed synchronization and frame structure, and implemented simultaneous real-time video, audio, and text transmission.
- Completed coding for audio and text transmissions and coordinated system integration.

PUBLICATIONS

- Yubo Zhang, Xiao-Yang Liu, and Xiaodong Wang, "Transformer-based Scalable Beamforming Optimization via Deep Residual Learning," ArXiv preprint ArXiv:2510.13077 (2025).
- Yubo Zhang, Jeremy Johnston, and Xiaodong Wang, "An Encoder-Decoder Network for Beamforming over Sparse Large-Scale MIMO Channels," ArXiv preprint ArXiv:2510.02355 (2025).
- Yubo Zhang, Hassan ZivariFard, and Xiaodong Wang, "Power and Rate Allocations for Positive-rate Covert Communications in Block-Fading Channels," IEEE Transactions on Wireless Communications (Revision).
- Yubo Zhang, Yuchen Pan, Chen Gong, Beiyuan Liu and Zhengyuan Xu, "Channel Estimation and Signal Detection for NLOS Ultraviolet Scattering Communication with Space Division Multiple Access," IEEE Transactions on Communications 72.10 (2024): 6427-6441.
- Yubo Zhang, Chen Gong, and Zhengyuan Xu, "Bi-directional ultra-violet communication with self-interference," Opt. Express 30 (21), 38534-38549 (2022).
- Yifan Ding, Yubo Zhang, Huabin Yu, Chen Gong, Haiding Sun, and Zhengyuan Xu, "266 nm ultraviolet communication under unknown interference using UVC micro-LED," Opt. Express 31.10 (2023): 16406-16422.
- Yuchen Pan, Yubo Zhang, Fei Long, Ping Li, Haotian Shi, Jiazhao Shi, Hanlin Xiao, Chen Gong, Zhengyuan Xu, "Beacon-enabled TDMA Ultraviolet Communication Network Design and Realization," CSNDSP 2024.
- Yuchen Pan, Fei Long, Yubo Zhang, Shihui Yu, Junfan Chen, Chen Gong, and Zhengyuan Xu, "Beacon-Based Time Synchronization Protocol Design for Layered Ultraviolet Communication Network," IEEE Transactions on Communications (2025).
- Yuchen Pan, Guanchu Wang, Yubo Zhang, Jingyin Tang, Chen Gong, and Zhengyuan Xu, "Graph-based conflict-free MAC protocol and conflict analysis for a two-layer ultraviolet communication network," JOCN 15.6 (2023): 381-392.
- Yifan Ding, Chen Gong, Weijie Liu, Yubo Zhang, Haiding Sun and Zhengyuan Xu, "Multi-Input UV Communication with Adaptive Gain Adjustment," Journal of Lightwave Technology, vol. 43, no. 3, pp. 1170-1178 (2025).

SKILLS & AWARDS

- Programming: Python (6 yrs), PyTorch (4 yrs), C (4 yrs), C++ (2 yrs), MATLAB (6 yrs), Verilog (4 yrs)
- Awards: G.D.C. Corporation Scholarship (2%), Outstanding Graduates of USTC (10%), Excellent Students of USTC, Silver Award (10%, for two years)
- Relevant courses: Optimization (100/100), Stochastic process (97/100), Optical network (97/100)